

Analysis Plan: One-Archetype Freeze-and-Stress Across Representative Köppen Climate Types

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1 Core Analytical Framing

This study uses a fixed medium-office archetype as a controlled test object to isolate the effect of future climate forcing on building resilience outcomes. Rather than comparing multiple building types, the analysis asks:

Holding building form, use, and HVAC system constant, how does future climate stress reorganize the dominant resilience failure mode across representative Köppen climate types?

The central claim is therefore not that all buildings respond identically, but that a fixed building-system configuration exhibits distinct climate-type-specific degradation profiles under future weather stress.

2 Primary Research Questions

1. Which literature-derived resilience metrics are most sensitive to future climate forcing for a fixed medium-office archetype?
2. Which climate-stressor blocks explain the observed changes in these metrics?
3. Do representative Köppen climate types produce distinct resilience failure profiles?
4. Which resilience family becomes dominant or breaches first under each climate profile?

3 Study Matrix

The proposed study matrix is:

6 representative Köppen climate types $\times$ 1 archetype $\times$ 2–3 climate pathways $\times$ 76 future years $\times$ 30 literature-derived metrics

where the fixed archetype is:

medium office

The basic annual panel structure is:

city $\times$ pathway $\times$ year $\times$ metric

If each Köppen class is represented by only one city, Köppen class should be treated primarily as a stratification and interpretation framework rather than as a fully replicated statistical factor.

4 Master Analysis Matrix

The main analysis table should contain one row per:

$$\text{city} \times \text{pathway} \times \text{year} \times \text{metric}$$

Recommended columns include:

- city
- koppen_class
- pathway
- year
- metric_name
- metric_family
- metric_value
- baseline_value
- delta_abs
- delta_pct
- delta_log
- delta_std
- breached
- driver_delta_Tmean
- driver_delta_CDD
- driver_delta_HDD
- driver_delta_WB
- driver_delta_GHI
- driver_heatwave_days
- driver_hot_nights
- driver_warm_spell_length

This table is the core analytical object for the paper.

5 Baseline Definition

Because the EnergyPlus experiment covers 2025–2100, the baseline should be defined using the first simulation decade:

$$\text{baseline} = 2025\text{--}2034$$

For each metric:

$$\Delta M_y = M_y - \bar{M}_{2025-2034}$$

A decadal baseline is preferred over a single-year baseline because it reduces sensitivity to annual variability.

6 Response Variables

Raw percentage change should not be used as the sole response because small baseline values can generate inflated percentage changes. Three response forms should be computed.

6.1 Absolute Change

$$\Delta M_{c,p,y,m} = M_{c,p,y,m} - \bar{M}_{c,p,\text{baseline},m}$$

This preserves physical units.

6.2 Log-Ratio Change

$$Y_{c,p,y,m} = \log(M_{c,p,y,m} + \epsilon) - \log(\bar{M}_{c,p,\text{baseline},m} + \epsilon)$$

This is useful for skewed metrics and avoids unstable percentage-change behavior.

6.3 Standardized Change

$$Z_{c,p,y,m} = \frac{M_{c,p,y,m} - \bar{M}_{c,p,\text{baseline},m}}{\sigma_{c,p,\text{baseline},m}}$$

For cross-metric comparison, standardized change or log-ratio change should be preferred. Absolute change should be retained for interpretation.

7 Climate-Stressor Predictors

The explanatory variables should capture more than annual mean temperature change. Recommended predictors include:

- annual mean dry-bulb temperature shift, ΔT_{mean}
- cooling degree-day change, ΔCDD
- heating degree-day change, ΔHDD
- wet-bulb or humidity-ratio shift, ΔWB
- global horizontal irradiance shift, ΔGHI
- heatwave-day count
- hot-night count
- warm-spell length or consecutive-hot-day count

Because ΔT_{mean} , ΔCDD , and ΔHDD are highly correlated, interpretation should focus on physically meaningful predictor blocks rather than individual coefficients.

8 Predictor Blocks

Predictors should be grouped into climate-stressor blocks.

8.1 Thermal Burden Block

$$B_{\text{thermal}} = \{\Delta T_{\text{mean}}, \Delta CDD, \Delta HDD, \text{hot nights}\}$$

8.2 Moisture Block

$$B_{\text{moisture}} = \{\Delta WB, \Delta q, \text{high-RH hours}\}$$

8.3 Solar Block

$$B_{\text{solar}} = \{\Delta GHI, \text{extreme solar hours}\}$$

8.4 Persistence Block

$$B_{\text{persistence}} = \{\text{heatwave days, consecutive hot days, warm-spell length}\}$$

Variance decomposition should then be performed at the block level.

9 Metric Registry

All candidate metrics should be literature-derived and registered before screening. The metric registry should include:

Metric	Family	Formula	Unit	Threshold	Source	E+ Variables
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Each metric should be assigned to one of the following families:

1. operational adequacy
2. habitability / human resilience
3. durability / moisture risk
4. valuation / secondary overlay

The core paper should emphasize the first three families. Valuation metrics such as carbon, cost, or regret can be used as secondary overlays but should not dominate the main analysis.

10 Metric Screening

All literature-derived metrics can be computed initially, but the final paper should not report all metrics equally. Each metric should be screened according to the following criteria.

10.1 Climate Sensitivity

$$S_1 = |\bar{Y}_{2071-2100} - \bar{Y}_{2025-2034}|$$

10.2 Temporal Trend

$$S_2 = |\beta_{\text{year}}|$$

where β_{year} is estimated from the annual or decadal time series.

10.3 Volatility Change

$$S_3 = SD(Y_{2071-2100}) - SD(Y_{2025-2034})$$

10.4 Breach Relevance

$$S_4 = \Pr(\text{breach by 2050 or 2100})$$

for metrics with defined thresholds.

10.5 Climate Discrimination

$$S_5 = \frac{\text{between-climate variance}}{\text{total variance}}$$

10.6 Redundancy Penalty

If two metrics are highly correlated across the annual panel, for example $r > 0.9$, one should be retained as the representative indicator.

The screening table should take the form:

Metric	Sensitivity	Trend	Volatility	Breach	Climate Disc.	Redundant?	Keep?
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11 Family-Level Scores

After screening, selected metrics can be combined into family-level scores. This reduces the 30-metric bank into interpretable resilience dimensions.

11.1 Operational Adequacy Score

$$O = z(\text{annual electricity}) + z(\text{peak demand}) + z(\text{unmet hours})$$

11.2 Habitability Score

$$H = z(\text{overheating degree-hours}) + z(\text{SETex}) + z(-\text{HoS})$$

where lower Hours of Safety implies higher risk.

11.3 Durability Score

$$D = z(\text{Mindex}) + z(\text{high-RH hours}) + z(\text{surface condensation risk})$$

The durability score should only be used if the EnergyPlus outputs support a defensible moisture-risk calculation. If only bulk zone relative humidity is available, durability should be reported cautiously or moved to supplementary analysis.

12 Decadal Aggregation

Annual trajectories should be plotted directly, but inferential analysis should also use decadal aggregation to avoid treating 76 annual observations as fully independent.

Suggested decadal bins are:

$$\begin{array}{cccc} 2025-2034, & 2035-2044, & 2045-2054, & 2055-2064, \\ 2065-2074, & 2075-2084, & 2085-2094, & 2095-2100 \end{array}$$

Each row in the decadal table becomes:

$$\text{city} \times \text{pathway} \times \text{decade}$$

for each metric or family score.

13 GAM / Mixed-Model Attribution

For each family score, a generalized additive model can be used:

$$Score_{k,c,p,y} = s(\Delta CDD) + s(\Delta HDD) + s(\Delta WB) + s(\Delta GHI) + s(\text{heatwave days}) + \alpha_{\text{pathway}} + u_{\text{city}} + \epsilon$$

where:

- $Score_k$ is one of O , H , or D ;
- $s(\cdot)$ denotes a smooth function;
- α_{pathway} captures pathway-level differences;

- u_{city} captures city-specific effects.

If only one city represents each Köppen class, the city random effect is effectively entangled with climate type. Therefore, Köppen should be used mainly for profile interpretation unless multiple cities per climate class are available.

A simpler alternative is a block-level model:

$$Score_k \sim B_{\text{thermal}} + B_{\text{moisture}} + B_{\text{solar}} + B_{\text{persistence}} + \text{pathway} + u_{\text{city}} + \epsilon$$

14 Variance Decomposition

The main statistical output should be block-level explained variance rather than individual coefficients.

14.1 Drop-One-Block Decomposition

Fit the full model and compute:

$$R_{\text{full}}^2$$

Then remove one predictor block at a time:

$$R_{-g}^2$$

The contribution of block g is:

$$C_g = R_{\text{full}}^2 - R_{-g}^2$$

where g may be thermal, moisture, solar, persistence, pathway, or city.

14.2 Grouped Permutation Importance

As a robustness check, permute each predictor block and compute the loss in predictive performance:

$$C_g = R_{\text{full}}^2 - R_{\text{permuted}(g)}^2$$

This provides an interpretable estimate of how much each climate-stressor block contributes to explaining resilience degradation.

15 Climate-Profile Classification

For each city, pathway, and time period, assign a dominant resilience profile.

A simple severity-based rule is:

$$\text{Dominant profile} = \arg \max(O, H, D)$$

A threshold-based rule is:

$$\text{First breached family} = \min(y_O^*, y_H^*, y_D^*)$$

where y_k^* is the first year in which family k crosses its breach threshold.

The final profile table should take the form:

Köppen Type	Dominant Failure Family	Main Driver Block	Adaptation Priority
Hot-humid	Habitability / latent stress	Moisture + hot nights	Passive survivability + dehumidification
Hot-dry	Operational peak stress	CDD + solar	Shading + cooling capacity
Cold continental	Cooling transition risk	Rising CDD + falling HDD	Cooling retrofit + controls
Marine	Mixed humidity stress	Moisture + mild overheating	Ventilation + envelope moisture control

16 Recommended Analysis Sequence

1. Build a metric registry for all literature-derived candidate metrics.
2. Confirm required EnergyPlus variables for each metric.
3. Generate the annual metric table for the fixed medium-office archetype.
4. Generate the annual climate-driver table for each city and pathway.
5. Join metric outcomes and climate drivers into the master panel.
6. Compute absolute, log-ratio, and standardized metric changes relative to the 2025–2034 baseline.
7. Screen metrics using sensitivity, trend, volatility, breach relevance, climate discrimination, and redundancy.
8. Select representative metrics per resilience family.
9. Construct operational, habitability, and durability family scores.
10. Fit GAM or mixed-effect attribution models for the family scores.
11. Perform block-level variance decomposition.
12. Classify each representative climate type by dominant failure family and main climate driver.
13. Conduct robustness checks across pathways and leave-one-city-out sensitivity.

17 Expected Figure Set

The paper should produce the following figures:

1. **Metric taxonomy:** literature-derived metrics grouped into resilience families.
2. **Climate-driver trajectories:** ΔT , ΔCDD , ΔHDD , ΔWB , ΔGHI , heatwave days, and hot nights by city/pathway.
3. **Metric-screening heatmap:** sensitivity, trend, volatility, breach frequency, climate discrimination, and redundancy for all candidate metrics.
4. **Family-score trajectories:** operational, habitability, and durability scores over 2025–2100.
5. **Variance-decomposition figure:** explained variance by thermal, moisture, solar, persistence, pathway, and city.
6. **Climate-profile matrix:** Köppen type $\rightarrow$ dominant failure family $\rightarrow$ main driver block $\rightarrow$ adaptation implication.

18 Core Contribution

The paper should not be framed as a generic screening of 30 metrics. The contribution should be:

Using a fixed medium-office archetype as a controlled test object, we show that future climate stress reorganizes building resilience risk differently across representative Köppen climate types.

More sharply:

Holding the building constant, future warming does not degrade performance along a single axis; climate type determines whether operational adequacy, habitability, or durability becomes the dominant failure mode.